

```

c.add( undr );
ButtonGroup bg = new ButtonGroup();
bg.add( ital );
bg.add( bold );
bg.add( undr );

```

The process of collecting JRadioButtons into a ButtonGroup is entirely separate from the creation of the GUI.

10.9 JMenuBar, JMenu, JMenuItem

The menu bar is an aspect of modern GUIs that users have become very common. The tool bar is just as common. Creating and managing either one requires significant effort. There are two broad categories of menus:

- Menu Bars (on the top) (Known as a top-level menu) (Menus that are contained in another menu are called pull-right menus.)
- Popup Menus (coming from anywhere else)

Menus can only be attached to objects of classes that contain the method:
`setJMenuBar()`

If you understand the terms used to describe a menu, building one can be very easy.

The bar across the top is the JMenuBar. After you instantiate the single JMenuBar, you set it to be the JMenuBar. To place a JMenu object in the JMenuBar, you add it. A JMenu item always has children. If this object is the end of the road, it is a JMenuItem . Because this has children menus, it is a JMenu .

Example:

```

JMenuBar bar = new JMenuBar();
setJMenuBar( bar );
action = new JMenu( "Action" );
enterNew = new JMenuItem( "Enter New Data" );
enterNew.addActionListener( this );
printData = new JMenu( "Print Data" );
printDataSorted = new JMenuItem( "Printed in Sorted
Order" );
printData.add( printDataSorted );
printDataEntry = new JMenuItem( "Print in Entry Order"
);
printData.add( printDataEntry );
exit = new JMenuItem( "Exit" );

```